



Jin-Hyeong Park

Machine Learning Engineer at AI Research Institute, Neowiz | Gyeonggi-do, South Korea

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Research Interests

Inverse Rendering, Neural 3D Representation, Neural Relighting

EDUCATION

Chung-Ang University (CAU), Seoul, Korea Mar 2018 - Feb 2020

M.S. in Computer Science and Engineering (Focus: Neural Architecture Search)

- Advisor: [Jaesung Lee](#) | GPA 3.82/4.0
- Thesis: Effective Front-end Architecture Search for Random Weight Network on Edge Device

Korea National University of Transportation (KNUT), Gyeonggi-do, South Korea Mar 2014 - Feb 2018

B.S. in Computer Science and Information Engineering

- Advisor: Sungwook Lee | GPA 3.65/4.0

PROFESSIONAL EXPERIENCE

AI Research Lab, Neowiz Nov 2020 - Present

Machine Learning Engineer

- Contributed to the development of the AI portion of an Apple's 2023 Mac Game of the Year Nominated AAA game title (Lies of P)
- Built and deployed diffusion-based image generation workflows and geometry-processing pipelines used across seven game projects.

Automated Machine Learning Lab, Chung-Ang University (CAU) | PI: Dr. Jaesung Lee

Graduate Research Assistant

Mar 2018 - Aug 2020

Research Intern

Jul 2017 - Feb 2018

PUBLICATIONS AND PRESENTATIONS

Decoupled Reprojection Consistency for Diagnosing 3D Gaussian Splatting Failures

Jin-Hyeong Park

Eurographics 2026 Poster

Multi-label Naïve Bayes Classifier Considering Label Dependence

Hae Cheon Kim, Jin-Hyeong Park, Dae Won Kim, Jaesung Lee

Pattern Recognition Letters, 2020

Compact Feature Subset Based Multi-label Music Categorization for Mobile Devices

Jaesung Lee, Wangduk Seo, Jin-Hyeong Park, Dae Won Kim

Multimedia Tools and Applications, 2018

Relevant Research & Industry Experience

Enhanced Game Image Generation AI Service Development and Feature Integration Jan 2024 - Present

Neowiz, South Korea

Keyword: Image processing, Generative Models, Game Art Generation, Human-Computer Interaction

- Developed Stable Diffusion-based features for concept art generation, style transfer, and background variation, enhancing the game artwork production workflow.

**Automated Hair Guide Model Generation for Game Characters
from 3D Reconstruction**

Apr 2024 - Nov 2024

Neowiz, South Korea

Keyword: 3D Content Generation, Geometry Processing, Inverse Rendering

- Designed a hair guide model generation pipeline to optimize the creation of game hair models, utilizing Geometry Processing techniques including clustering, importance sampling, and resampling.
- Developed a post-processing procedure that parametrically differentiates guide hair using results from existing Inverse Rendering research.

Development of Facial Animation Pipeline for Lip Sync and Emotion based on Voice and Script [\[video\]](#)

Aug 2022 - Oct 2022

Neowiz, South Korea

Keyword: Signal Processing, Multivariate Prediction, Sentiment Analysis

- Designed and implemented an automated pipeline for facial animations in cartoon-style games, utilizing script-based sentiment analysis and speech-to-viseme models to automate facial expression generation

Neural Audio Filter for Transforming Monster Voices into Machinery Sounds [\[video\]](#)

Jun 2021 - Oct 2022

Neowiz, South Korea

Keyword: Neural Representation, Signal Processing, GAN-based Style Transfer

- Conducted GAN-based domain transfer to convert monster voices into mechanical sounds; applied in the AAA game Lies of P (Apple's 2023 Mac Game of the Year).

Development of Computer-based Three-Dimensional Medical Image Analysis Program for the Objective Assessment of Orbital Disease

Sep 2018 - Oct 2019

National Research Foundation of Korea

Keyword: Computer Vision, 3D CNNs

- Conducted 3D volumetric data classification using 3D convolutional neural networks, achieving high accuracy in multi-class prediction tasks.

SKILLS

Languages: Python • C++ • CUDA • Javascript • LaTeX

Framework: PyTorch • DirectX 11 • Flutter

Software: Blender • COLMAP • Unreal Engine • ComfyUI • Git • MATLAB

AWARDS AND HONORS

Funding & Scholarships

Research Assistantship (A), CAU

Mar 2018 - Mar 2020

- Full Tuition Scholarship

National Science & Technology Excellence Undergraduate Scholarship, KSF

Sep 2016 - Jun 2017

- Full Tuition Scholarship

Academic Excellence Scholarship, KNUT

- Half Tuition Scholarship
- Half Tuition Scholarship

Mar 2016 - Jun 2016

Sep 2014 - Dec 2014

Awards

- **Bronze Medal**, CommonLit Readability Prize, *Kaggle* (Top 10%) 2021
- **Silver Medal**, Ion Switching Prediction, *Kaggle* (Top 4%) 2020

TEACHING EXPERIENCE

Teaching Mentor for Undergraduate Interns, CAU

Sep 2018 - Mar 2020

- Mentored experimental design and academic paper composition

Teaching Assistant, CAU

Artificial Intelligence Class

Mar 2019 - May 2019

Numerical Analysis Class

Sep 2018 - Oct 2018

Discrete Mathematics Class

Mar 2018 - May 2018

MEMBERSHIP

- **Association for Computing Machinery (ACM):** Professional Member (2024–)
- **BrainKorea(BK)21 Four:** Graduate Research Program Member (2018–).

EXTRA EXPERIENCE

- Lab Manager for 2 years (2018-2019), managed primary lab operations and coordinating research activities at the AutoML Lab, CAU